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## Power supply switch circuit and operating method thereof


## Abstract

A power supply switch circuit includes: a first switch to switch supply of a first power supply voltage to a power supply terminal of a power amplifier; a control voltage generator to generate a first control voltage through a charge pump operation, and to control the charge pump operation by comparing the first power supply voltage to a first voltage, which is a voltage of the power supply terminal; and a switch controller to generate a switching driving signal to control the first switch using the first control voltage.



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## Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS
(1) This application claims priority to and the benefit under 35 USC 119(a) of Korean Patent Application No. 10-2021-0135175 filed in the Korean Intellectual Property Office on Oct. 12, 2021, the entire contents of which are incorporated herein by reference for all purposes.
BACKGROUND
Field
(2) The following description relates to a power supply switch circuit and an operation method thereof.
Description of the Background
(3) As wireless communication standards evolve, a plurality of communication standards such as 2G, Wi-Fi, Bluetooth, 3G, 4G, and 5G are used in a single device (e.g., a smartphone). As a plurality of communication standards are used in one device, a power amplifier that outputs a transmission signal is also used for each communication standard. That is, in order to output a signal conforming to the plurality of communication standards, a plurality of power amplifiers corresponding to the plurality of communication standards may be required.
(4) The power amplifier operates by receiving power supply from the outside, and in general, a separate power supply integrated circuit (IC) that supplies power to a single power amplifier is used. For example, four power supply ICs are used to operate four power amplifiers. When one communication standard among the plurality of communication standards is used, the other

communication standards may not be simultaneously used. For example, when the 4G communication standard is used, the 3G communication standard may not be used. Accordingly, a power supply IC corresponding to an unused communication standard needs to be effectively used for another communication standard.

SUMMARY

(5) This Summary is provided to introduce a selection of concepts in simplified form that are further described below in the Detailed Description. This Summary is not intended to identify key features or essential features of the claimed subject matter, nor is it intended to be used as an aid in determining the scope of the claimed subject matter.

(6) In one general aspect, a power supply switch circuit includes: a first switch configured to switch supply of a first power supply voltage to a power supply terminal of a power amplifier; a control voltage generator configured to generate a first control voltage through a charge pump operation, and to control the charge pump operation by comparing the first power supply voltage to a first voltage, which is a voltage of the power supply terminal; and a switch controller configured to generate a switching driving signal to control the first switch using the first control voltage.

(7) The power supply switch circuit may include a second switch configured to switch supply of a second power supply voltage to the power supply terminal. The control voltage generator may be configured to generate a second control voltage through a second charge pump operation, and to control the second charge pump operation by comparing the second power supply voltage to the first voltage, and the switch controller may be configured to generate a second switching driving signal to control the second switch using the second control voltage.

(8) The control voltage generator may include: a comparator configured to compare the first power supply voltage to the first voltage; an oscillator configured to generate a waveform corresponding to an enable signal of the first switch and an output of the comparator; and a charge pump configured to receive the first power supply voltage and a second voltage, and to generate the first control voltage by operating with the waveform.

(9) The power supply switch circuit may include a counter configured to count a number of clocks of the waveform, and to output an enable signal to the comparator when the number of clocks satisfies a reference number. The comparator may be configured to determine whether a difference between the first power supply voltage and the first voltage is included within a set range when the enable signal of the counter is input.

(10) The oscillator may be configured to stop generating the waveform when the comparator determines that the difference between the first power supply voltage and the first voltage is included within the set range.

(11) The control voltage generator may include a NAND gate configured to receive the enable signal of the first switch and an output of the comparator, and the oscillator may be activated or deactivated corresponding to an output signal of the NAND gate.

(12) The first control voltage may correspond to a sum of the first power supply voltage and the second voltage.

(13) The switch controller may include a buffer circuit configured to receive the first control voltage and to generate the switching driving signal having the first control voltage.

(14) The first power supply voltage may be fluctuated according to an envelope of a radio signal (RF) input to the power amplifier.

(15) In another general aspect, an operation method of a power supply switch circuit that supplies at least one of a first power supply voltage and a second power supply voltage to a power supply terminal of a power amplifier by switching the at least one voltage includes: generating a first waveform corresponding to an enable signal with respect to a first switch that switches the first power supply voltage; generating a first control voltage by performing a charge pump operation through the first waveform; comparing the first power supply voltage to a first voltage, which is a voltage of the power supply terminal; stopping the generating of the first waveform corresponding

to a result of the comparing; and generating a first switching driving signal that controls the first switch using the first control voltage.
(16) The method may include counting a number of clocks with respect to the first waveform, and the comparing may include comparing the first power supply voltage to the first voltage when the number of clocks satisfies a reference number.
(17) The stopping may include stopping generating the first waveform when a difference between the first power supply voltage and the first voltage is included within a set range.
(18) The method may include: generating a second waveform corresponding to an enable signal with respect to a second switch that switches the second power supply voltage; generating a second control voltage by performing a second charge pump operation through the second waveform; comparing the second power supply voltage to the first voltage; stopping generating the second waveform corresponding to a result of the comparing between the second power supply voltage and the first voltage; and generating a second switching driving signal that controls the second switch using the second control voltage.
(19) The first control voltage may be higher than the first power supply voltage, and the second control voltage may be higher than the second power supply voltage.
(20) The first power supply voltage and the second power supply voltage may be fluctuated according to an envelope of a radio frequency (RF) signal input to the power amplifier.
(21) Other features and aspects will be apparent from the following detailed description, the drawings, and the claims.

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## Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1** is a block diagram of a transmission system according to an example.
(2) FIG. **2** shows a connection relationship between the power supply switch circuit and the power amplifier according to an example.
(3) FIG. **3** shows an internal configuration of the power supply switch circuit of FIG. **2**.
(4) FIG. **4** shows a logic table according to an example.
(5) FIG. **5** shows the control voltage generator according to an example.
(6) FIG. **6**A shows an internal configuration of the first control voltage generator according to an example.
(7) FIG. **6**B shows an internal configuration of the second control voltage generator according to an example.
(8) FIG. **7** shows an operation timing graph with respect to the first control voltage generator of FIG. **6**A.
(9) FIG. **8** shows an internal configuration of the switch circuit and an internal configuration of the switch controller according to an example.
(10) FIG. **9** shows the logic circuit according to an example.
(11) FIG. **10** shows an input-output logic table of the logic circuit of FIG. **9**.
(12) Throughout the drawings and the detailed description, the same reference numerals refer to the same elements. The drawings may not be to scale, and the relative size, proportions, and depictions of elements in the drawings may be exaggerated for clarity, illustration, and convenience.
DETAILED DESCRIPTION
(13) The following detailed description is provided to assist the reader in gaining a comprehensive understanding of the methods, apparatuses, and/or systems described herein. However, various changes, modifications, and equivalents of the methods, apparatuses, and/or systems described herein will be apparent to one of ordinary skill in the art. The sequences of operations described herein are merely examples, and are not limited to those set forth herein, but may be changed as

will be apparent to one of ordinary skill in the art, with the exception of operations necessarily occurring in a certain order. Also, descriptions of functions and constructions that would be well known to one of ordinary skill in the art may be omitted for increased clarity and conciseness.

(14) The features described herein may be embodied in different forms, and are not to be construed as being limited to the examples described herein. Rather, the examples described herein have been provided so that this disclosure will be thorough and complete, and will fully convey the scope of the disclosure to one of ordinary skill in the art.

(15) Herein, it is noted that use of the term “may” with respect to an example or embodiment, e.g., as to what an example or embodiment may include or implement, means that at least one example or embodiment exists in which such a feature is included or implemented while all examples and embodiments are not limited thereto.

(16) Throughout the specification, when an element, such as a layer, region, or substrate, is described as being “on,” “connected to,” or “coupled to” another element, it may be directly “on,” “connected to,” or “coupled to” the other element, or there may be one or more other elements intervening therebetween. In contrast, when an element is described as being “directly on,” “directly connected to,” or “directly coupled to” another element, there can be no other elements intervening therebetween.

(17) As used herein, the term “and/or” includes any one and any combination of any two or more of the associated listed items.

(18) Although terms such as “first,” “second,” and “third” may be used herein to describe various members, components, regions, layers, or sections, these members, components, regions, layers, or sections are not to be limited by these terms. Rather, these terms are only used to distinguish one member, component, region, layer, or section from another member, component, region, layer, or section. Thus, a first member, component, region, layer, or section referred to in examples described herein may also be referred to as a second member, component, region, layer, or section without departing from the teachings of the examples.

(19) Spatially relative terms such as “above,” “upper,” “below,” and “lower” may be used herein for ease of description to describe one element's relationship to another element as illustrated in the figures. Such spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. For example, if the device in the figures is turned over, an element described as being “above” or “upper” relative to another element will then be “below” or “lower” relative to the other element. Thus, the term “above” encompasses both the above and below orientations depending on the spatial orientation of the device. The device may also be oriented in other ways (for example, rotated 90 degrees or at other orientations), and the spatially relative terms used herein are to be interpreted accordingly.

(20) The terminology used herein is for describing various examples only, and is not to be used to limit the disclosure. The articles “a,” “an,” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. The terms “comprises,” “includes,” and “has” specify the presence of stated features, numbers, operations, members, elements, and/or combinations thereof, but do not preclude the presence or addition of one or more other features, numbers, operations, members, elements, and/or combinations thereof.

(21) Due to manufacturing techniques and/or tolerances, variations of the shapes illustrated in the drawings may occur. Thus, the examples described herein are not limited to the specific shapes illustrated in the drawings, but include changes in shape that occur during manufacturing.

(22) The features of the examples described herein may be combined in various ways as will be apparent after an understanding of the disclosure of this application. Further, although the examples described herein have a variety of configurations, other configurations are possible as will be apparent after an understanding of the disclosure of this application.

(23) The drawings may not be to scale, and the relative sizes, proportions, and depiction of elements in the drawings may be exaggerated for clarity, illustration, and convenience.

(24) Throughout the description, RF signals may have a format according to Wi-Fi (IEEE 802.11 family, etc.), WiMAX (IEEE 802.16 family, etc.), IEEE 802.20, LTE (long term evolution), EV-DO, HSPA, HSDPA, HSUPA, EDGE, GSM, GPS, GPRS, CDMA, TDMA, DECT, Bluetooth, 3G, 4G, 5G, and any other wireless and wired protocols designated thereafter, but is not limited thereto.
(25) FIG. **1** is a block diagram of a transmission system **1000** according to an example.
(26) As shown in FIG. **1**, the transmission system **1000** may include first and second power supply circuits **100***a* and **100***b*, first and second power supply switch circuits **200***a* and **200***b*, and first, second, third, and fourth power amplifiers **300***a*, **300***b*, **300***c*, and **300***d*. FIG. **1** illustrates that the transmission system **100** is formed of four power amplifiers, and two power supply switch circuits are connected to the four power amplifiers, but the configuration is not limited thereto and the number of power amplifiers and the number of power supply switch circuits can be changed.
(27) The first power supply circuit **100***a* generates and outputs a first power supply voltage VCC**1**. The first power supply voltage VCC**1** may be applied to a power supply terminal of the first power amplifier **300***a*, the second power amplifier **300***b*, or the third power amplifier **300***c*. In order to support an average power tracking mode (APT mode), a value of the first power supply voltage VCC**1** may be changed according to an envelope of a signal input to the first power amplifier **300***a*, the second power amplifier **300***b*, or the third power amplifier **300***c*.
(28) The second power supply circuit **100***b* generates and outputs a second power supply voltage VCC**2**. The second power supply voltage VCC**2** may be applied to a power supply terminal of the second power amplifier **300***b*, the third power amplifier **300***c*, or the fourth power amplifier **300***d*. In order to support the APT mode, a value of the second power supply voltage VCC**2** may be changed according to an envelope of a signal input to the second power amplifier **300***b*, the third power amplifier **300***c*, or the fourth power amplifier **300***d*.
(29) Each of the first and second power supply circuits **100***a* and **100***b* may be implemented as a power management (PM) IC.
(30) The first power supply switch circuit **200***a* may receive the first power supply voltage VCC**1** from the first power supply circuit **100***a* and may receive the second power supply voltage VCC**2** from the second power supply circuit **100***b*. The first power supply switch circuit **200***a* may select one of the input first and second power supply voltages VCC**1** and VCC**2** and output the selected one to the power supply terminal of the second power amplifier **300***b*. For example, when the first power amplifier **300***a* does not operate, the first power supply switch circuit **200***a* may select the first power supply voltage VCC**1** and output the first power supply voltage VCC**1** to the power supply terminal of the second power amplifier **300***b*. In addition, when the fourth power amplifier **300***d* does not operate, the first power supply switch circuit **200***a* may select the second power supply voltage VCC**2** and output the second power supply voltage VCC**2** to the power supply terminal of the second power amplifier **300***b*.
(31) The second power supply switch circuit **200***b* receives the first power supply voltage VCC**1** from the first power supply circuit **100***a* and receives the second power supply voltage VCC**2** from the second power supply circuit **100***b*. The second power supply switch circuit **200***b* may select one of the input first and second power supply voltages VCC**1** and VCC**2** and output the selected one to the power supply terminal of the third power amplifier **300***c*. For example, when the first power amplifier **300***a* does not operate, the second power supply switch circuit **200***b* may select the first power supply voltage VCC**1** and output the first power supply voltage VCC**1** to the power supply terminal of the third power amplifier **300***c*. In addition, when the fourth power amplifier **300***d* does not operate, the second power supply switch circuit **200***b* may select the second power supply voltage VCC**2** and output the second power supply voltage VCC**2** to the power supply terminal of the third power amplifier **300***c*.
(32) The first power amplifier **300***a* operates by receiving the first power supply voltage VCC**1** from the first power supply circuit **100***a*, and amplifies and outputs an input radio frequency (RF) signal. The input RF signal of the first power amplifier **300***a* may be an RF signal for a first

communication standard.
(33) The second power amplifier **300***b* operates by receiving the power supply voltage (i.e., the first power supply voltage VCC**1** or the second power supply voltage VCC**2**) selected by the first power supply switch circuit **200***a*, and amplifies and outputs an input RF signal. The input RF signal of the second power amplifier **300***b* may be an RF signal for a second communication standard.
(34) The third power amplifier **300***c* operates by receiving the power supply voltage (i.e., the first power supply voltage VCC**1** or the second power supply voltage VCC**2**) selected by the second power supply switch circuit **200***b*, and amplifies and outputs the input RF signal. The input RF signal of the third power amplifier **300***c* may be an RF signal for a third communication standard.
(35) The fourth power amplifier **300***d* operates by receiving the second power supply voltage VCC**2** supplied from the second power supply circuit **100***b*, and amplifies and outputs an input RF signal. The input RF signal of the fourth power amplifier **300***d* may be an RF signal for a fourth communication standard.
(36) The first to fourth communication standards may be different from each other, and may be any one of 2G, WiFi, Bluetooth, 3G, 4G, and 5G communication standards. The first to fourth communication standards may be communication standards that define different bands among 5G communication standards.
(37) According to such a configuration, the number of power supply circuits can be reduced by sharing the power supply voltage through the power supply switch circuit. In general, when there are four power amplifiers, four power supply circuits are used, but in FIG. **1**, the number of power supply circuits can be reduced to two by using a power supply switch circuit. Hereinafter, a detailed configuration and operation method of a power supply switch circuit such as the first and second power supply switch circuits **200***a* and **200***b* will be described.
(38) FIG. **2** shows a connection relationship between the power supply switch circuit **200** and the power amplifier **300** according to the example.
(39) The power supply switch circuit **200** receives the first power supply voltage VCC**1** and the second power supply voltage VCC**2**, and selects and outputs one of the first and second power supply voltages VCC**1** and VCC**2** to a power supply terminal T_VCC of the power amplifier **300**. The power supply switch circuit **200** may be the first power supply switch circuit **200***a* or the second power supply switch circuit **200***b* of FIG. **1**.
(40) In FIG. **1** and FIG. **2**, the power supply switch circuit **200** receives two power supply voltages, but the power supply switch circuit **200** may receive at least two power supply voltages. In this case, the power supply switch circuit **200** may select one power supply voltage among at least two power supply voltages.
(41) The power amplifier **300** includes an input terminal RF.sub.in, an output terminal R.sub.out, and a power supply terminal T_VCC. The RF signal is input to the input terminal RF.sub.out, and the amplified signal is output to the output terminal RF.sub.out. A power supply voltage (VCC**1** or VCC**2**) is applied to the power supply terminal T_VCC, and the power amplifier **300** is operated by the applied power supply voltage (VCC**1** or VCC**2**). The power amplifier **300** may be implemented as a transistor. When the power amplifier **300** is implemented as a bipolar junction transistor (BJT), the input terminal RF.sub.in may be a base and the power supply terminal T_VCC may be a collector or an emitter. When the power amplifier **300** is implemented as a field effect transistor (FET), the input terminal RF.sub.in may be a gate and the power supply terminal T_VCC may be a drain or a source.
(42) The power supply switch circuit **200** and the power amplifier **300** of FIG. **2** may be combined to implement a single power amplifier module.
(43) FIG. **3** shows an internal configuration of the power supply switch circuit **200** of FIG. **2**.
(44) As shown in FIG. **3**, the power supply switch circuit **200** may include a switch circuit **210**, a switch controller **220**, and a control voltage generator **230**.
(45) The switch circuit **210** may include a first switch SW**1** and a second switch SW**2**. The first

switch SW**1** may switch the supply of the first power supply voltage VCC**1** to the power supply terminal T_VCC of the power amplifier **300**, and the second switch SW**2** may switch the supply of the second power supply voltage VCC**2** to the power supply terminal T_VCC of the power amplifier **300**. The first switch SW**1** may be connected between the first power supply circuit **100***a* and the power supply terminal T_VCC of the power amplifier **300**, and the second switch SW**2** may be connected between the second power supply circuit **100***b* and the power supply terminal T_VCC of the power amplifier **300**. In FIG. **3**, a voltage of the power supply terminal T_VCC is denoted as V.sub.T, and hereinafter, the voltage of the power supply terminal T_VCC will be called a power supply terminal voltage V.sub.T.
(46) The switch controller **220** may receive a bit signal (digital signal) from the outside and may generate a switching driving signal V.sub.SW for switching the switch circuit **210** in response to the input bit signal. The generated switching driving signal V.sub.SW is output to the switch circuit **210**. Here, as an example, a bit signal input from the outside may be 2 bits. The switching driving signal V.sub.SW may include a first switching driving signal V.sub.SW1 controlling the first switch SW**1** and a second switching driving signal V.sub.SW2 controlling the second switch SW**2**. The switch controller **220** receives the first control voltage V.sub.C1 from the control voltage generator **230** as an input, and may generate the first switching driving signal V.sub.SW1 by using the first control voltage V.sub.C1. In addition, the switch controller **220** receives a second control voltage V.sub.C2 from the control voltage generator **230** as an input, and may generate the second switching driving signal V.sub.SW2 by using the second control voltage V.sub.C2.
(47) When the first switching driving signal V.sub.SW1 is an ON driving signal and the second switching driving signal V.sub.SW2 is an OFF driving signal, the first switch SW**1** is turned on and the second switch SW**2** is turned off. Accordingly, the first power supply voltage VCC**1** is applied to the power supply terminal T_VCC of the power amplifier **300** through the first switch SW**1**.
(48) When the first switching driving signal V.sub.SW1 is an OFF driving signal and the second switching driving signal V.sub.SW2 is an ON driving signal, the first switch SW**1** is turned off and the second switch SW**2** is turned on. Accordingly, the second power supply voltage VCC**2** is applied to the power supply terminal T_VCC of the power amplifier **300** through the second switch SW**2**.
(49) FIG. **4** shows a logic table according to the example.
(50) In FIG. **4**, bit**1** and bit**2** denote external bit signals input to the switch controller **220**. As shown in FIG. **4**, when the external bit signal is 00 and 11, both the first and second switching driving signals V.sub.SW1 and V.sub.SW2 may be OFF driving signals, and the first and second switches SW**1** and SW**2** may both be in a turn-off state. The switch controller **220** may include a logic circuit having a logic table such as shown in FIG. **4**, which will be described in more detail below.
(51) As shown in FIG. **3**, the control voltage generator **230** receives the first power supply voltage VCC**1** from the first power supply circuit **100***a*, and receives the second power supply voltage VCC**2** from the second power supply circuit **100***b*. In addition, the control voltage generator **230** receives a first excessive voltage ΔV.sub.1, a second excessive voltage ΔV.sub.2, and a power supply terminal voltage V.sub.1. The control voltage generator **230** generates the first control voltage V.sub.C1 by using the first power supply voltage VCC**1**, the first excessive voltage ΔV.sub.1, and the power supply terminal voltage V.sub.1. The control voltage generator **230** generates the second control voltage V.sub.C2 by using the second power supply voltage VCC**2**, the second excessive voltage ΔV.sub.2, and the power supply terminal voltage V.sub.1. The first and second control voltages V.sub.C1 and V.sub.C2 generated by the control voltage generator **230** are input to the switch controller **220**. The first control voltage V.sub.C1 may be a control voltage (e.g., a gate voltage) for switching the first switch SW**1**, and the second control voltage V.sub.C2 may be a control voltage (e.g., a gate voltage) for switching the second switch SW**2**.
(52) FIG. **5** shows the control voltage generator **230** according to the example.
(53) As shown in FIG. **5**, the control voltage generator **230** may include a first control voltage

generator **230***a* and a second control voltage generator **230***b*.
(54) The first control voltage generator **230***a* receives the first power supply voltage VCC**1**, the first excessive voltage ΔV.sub.1, and the power supply terminal voltage V.sub.1, and generates the first control voltage V.sub.C1 using the received voltages. In more detail, the first control voltage generator **230***a* generates the first control voltage V.sub.C1 by operating an internal charge pump when an enable signal of the first switch SW**1** is input. In addition, the first control voltage generator **230***a* may reduce current consumption by operating the charge pump on an internal counter for a predetermined time. Here, the first control voltage V.sub.C1 may be the sum (VCC**1**+ΔV.sub.1) of the first power supply voltage VCC**1** and the first excessive voltage ΔV.sub.1. The first excessive voltage ΔV.sub.1 is an arbitrarily set voltage, and may be 3 V as an example. The first excessive voltage ΔV.sub.1 may be implemented through a regulator such as a low dropout (LDO).
(55) The second control voltage generator **230***b* receives the second power supply voltage VCC**2**, the second excessive voltage ΔV.sub.2, and the power supply terminal voltage V.sub.1, and generates a second control voltage V.sub.C2 using the received voltages. In more detail, the second control voltage generator **230***b* generates the second control voltage V.sub.C2 by operating the internal charge pump when an enable signal of the second switch SW**2** is input. In addition, the second control voltage generator **230***b* may reduce current consumption by operating the charge pump on an internal counter for a predetermined time. Here, the second control voltage V.sub.C2 may be the sum (VCC**2**+ΔV.sub.2) of the second power supply voltage VCC**2** and the second excessive voltage ΔV.sub.2. The second excessive voltage ΔV.sub.2 is an arbitrarily set voltage, and may be 3 V as an example. The second excessive voltage ΔV.sub.2 may also be implemented through a regulator such as an LDO. The second excessive voltage ΔV.sub.2 may be the same as the first excessive voltage ΔV.sub.1.
(56) FIG. **6**A shows an internal configuration of the first control voltage generator **230***a* according to the example, and FIG. **6**B shows an internal configuration of the second control voltage generator **230***b* according to the example.
(57) As shown in FIG. **6**A, the first control voltage generator **230***a* may include an oscillator **231***a*, a charge pump **232***a*, a counter **233***a*, and a comparator **234***a*.
(58) The oscillator **231***a* is activated when an enable signal SE**1**_Enable of the first switch is input, and may generate a square wave. The square wave generated by the oscillator **231***a* may be a clock signal. That is, when a turn-on signal of the first switch SW**1** is input, the oscillator **231***a* may be activated to generate a square wave. Here, the enable signal SW**1**_Enable of the first switch may be bit **1**, which is a bit signal for controlling the first switch SW**1**. In FIG. **6**A, an output signal of the oscillator **231***a* is denoted as “OSC_OUT”. On the other hand, as described below, the oscillator **231***a* may be deactivated in response to an output of the comparator **234***a* and stop generating the square wave.
(59) The charge pump **232***a* receives an output signal OSC_OUT of the oscillator **231***a* and receives the first power supply voltage VCC**1** and the first excessive voltage ΔV.sub.1. When a square wave is input from the oscillator **231***a*, the charge pump **232***a* operates. Accordingly, the charge pump **232***a* outputs a voltage that corresponds to the sum (VCC**1**+ΔV.sub.1) of the first power supply voltage VCC**1** and the first excessive voltage ΔV.sub.1 as the first control voltage V.sub.C1. Here, the charge pump **232***a* continues the charge pumping operation until a power terminal voltage (VT) becomes a voltage similar to the first power supply voltage VCC**1**. A method for the charge pump **233***a* to generate the first control voltage V.sub.C1 using the first power supply voltage VCC**1**, the first excessive voltage ΔV.sub.1, and the square wave input from the oscillator **231***a* may be a conventional method.
(60) Here, a relationship between the first control voltage V.sub.C1 and the first power supply voltage VCC**1** may satisfy the following Equation 1.
$V$.sub.C1=$VCC$1+Δ$V$.sub.1 (Equation 1)

(61) As shown in Equation 1, the first control voltage V.sub.C1 may be set as a higher voltage than the first power supply voltage VCC**1** by the first excessive voltage ΔV.sub.1.
(62) The first control voltage V.sub.C1 is a voltage used to switch the first switch SW**1**, and thus the first switch SW**1** may be sufficiently turned on. For example, when the first switch SW**1** is implemented as an N-type transistor, the voltage of the first switching driving signal V.sub.SW1 must be set higher than the first power supply voltage VCC**1** such that the first switch SW**1** can sufficiently perform a turn-on operation. Accordingly, the first control voltage generator **230***a* according to the embodiment generates a first control voltage V.sub.ci that is higher than the first power supply voltage VCC**1** and outputs it to the switch controller **220**.
(63) The counter **233***a* receives an output signal OSC_OUT of the oscillator **231***a* and performs counting. That is, when the oscillator **231***a* outputs a square wave, the counter **233***a* outputs the number of times the square wave is clocked. A predetermined reference number is set inside the counter **233***a*. The counter **233***a* outputs an enable signal when the counted number of clocks becomes a predetermined reference number. In addition, the counter **233***a* outputs a disable signal until the counted clocks reach the predetermined reference number. Here, the predetermined reference number is a number preset by a designer, and may be set corresponding to the number of pumping times at which the charge pump **233***a* can sufficiently generate the first control voltage. For example, when the designer knows that the first control voltage V.sub.C1 is sufficiently boosted with only 10 pumping operations of the charge pump **233***a*, the predetermined number of times can be set to 10.
(64) The comparator **234***a* receives the output signal of the counter **233***a* as an input and performs an operation. When an enable signal is input from the counter **233***a*, the comparator **234***a* is activated to perform a comparison operation. In addition, when a disable signal is input from the counter **233***a*, the comparator **234***a* is deactivated and a comparison operation is not performed. The first power supply voltage VCC**1** and the power terminal voltage V.sub.T to be compared are input to the comparator **234***a*, and the comparator **234***a* determines whether the first power supply voltage VCC**1** and the power terminal voltage V.sub.T are similar to each other. When the difference between the first power supply voltage VCC**1** and the power terminal voltage V.sub.T is within a predetermined range, the comparator **234***a* may determine them to be similar to each other. When the first power supply voltage VCC**1** and the power terminal voltage V.sub.T are similar to each other, the comparator **234***a* outputs a high signal. When the first power supply voltage VCC**1** and the power terminal voltage V.sub.T are not similar to each other, the comparator **234***a* outputs a low signal. When the first power supply voltage VCC**1** and the power terminal voltage V.sub.T are similar to each other, it means that the first power supply voltage VCC**1** is normally supplied to the power terminal T_VCC. A method for the comparator **234***a* to determine whether the two voltages VCC**1** and V.sub.T are similar may be a conventional method.
(65) The first control voltage generator **230***a* may further include a NAND gate **235***a*. The NAND gate **235***a* receives an enable signal SW**1**_Enable of the first switch and an output signal of the comparator **234***a*. An output signal of the NAND gate **235***a* is output to the oscillator **231***a*, and the oscillator **231***a* is activated corresponding to the output signal of the NAND gate **235***a*. When the output signal of the NAND gate **235***a* is a high signal, the oscillator **231***a* is activated. When the output signal of the NAND gate **235***a* is a low signal, the oscillator **231***a* is deactivated. The NAND gate **235***a* outputs a low signal only when both input signals (i.e., the enable signal SW**1**_Enable of the first switch and the output signal of the comparator **234***a*) are high signals. As described above, the comparator **234***a* outputs a high signal when the two voltages VCC**1** and V.sub.T are similar. Accordingly, when the two voltages VCC**1** and V.sub.T are similar to each other, the NAND gate **235***a* outputs a low signal, and the oscillator **231***a* is deactivated to stop generating a square wave. When the oscillator **231***a* stops generating a square wave, the charge pump **232***a* stops pumping.
(66) FIG. **7** shows an operation timing graph with respect to the first control voltage generator **230***a* of FIG. **6**A.

(67) At t**1**, the enable signal SW**1**_Enable of the first switch is changed from a low signal to a high signal. Accordingly, the oscillator **231**a is activated and generates a square wave. The square wave (e.g., a clock signal) generated by the oscillator **231**a is input to the charge pump **232**a and the counter **233**a. In this time, the charge pump **232**a performs a pumping operation and the counter **233**a performs a counting operation. The charge pump **233**a generates the first control voltage V.sub.C1 that rises through the pumping operation, and the counter **233**a counts the number of square waves of the oscillator **231**a.
(68) At time t**2**, the counter **233**a outputs an enable signal. That is, the counter **233**a outputs an enable signal when the counted number of clocks becomes a predetermined reference number. In this case, since the enable signal of the counter **233**a is input to the comparator **234**a, the comparator **234**a is activated to perform a comparison operation. Meanwhile, between the time t**1** and the time t**2**, the power terminal voltage V.sub.T gradually increases to reach near the first power supply voltage VCC**1**. At the time t**2**, the comparator **234**a determines whether the first power supply voltage VCC**1** and the power terminal voltage V.sub.T are similar to each other, and when they are similar, the comparator **234**a outputs a high signal. The NAND gate **235**a outputs a low signal by the high signal of comparator **234**a, and the oscillator **231**a is deactivated. Accordingly, at the time t**2**, the oscillator **231**a stops generating the square wave.
(69) In addition, the charge pump **232**a continues the pumping operation from the time t**1** and stops the pumping operation at the time t**2**. Since the charge pump **232**a has performed a sufficient pumping operation before the time t**2**, the first control voltage V.sub.C1 may be a voltage (VCC**1**+ΔV.sub.1) corresponding to the sum of the first power supply voltage VCC**1** and the first excessive voltage ΔV.sub.1. Since the first control voltage V.sub.C1 is used as a control voltage to turn on the first switch SW**1**, the first switch SW**1** can be sufficiently turned on. Accordingly, a voltage substantially similar to the first power supply voltage VCC**1** may be applied to the power terminal T_VCC.
(70) As shown in FIG. **6**B, a second control voltage generator **230**b according to an example may include an oscillator **231**b, a charge pump **232**b, a counter **233**b, a comparator **234**b, and a NAND gate **235**b.
(71) Referring to FIG. **6**A and FIG. **6**B, an internal configuration of the second control voltage generator **230**b is substantially to the same as that of the first control voltage generator **230**a except for the following parts, and detailed description thereof will be omitted. The oscillator **231**a generates a square wave when an enable signal SW**2**_Enable of the second switch is input. The charge pump **232**b receives a second power supply voltage VCC**2** and a second excessive voltage ΔV.sub.2, and outputs a voltage VCC**2**+ΔV.sub.2 corresponding to the sum of the second power supply voltage VCC**2** and the second excessive voltage ΔV.sub.2 as the second control voltage V.sub.C2. The second power supply voltage VCC**2** and the power terminal voltage V.sub.T are input to the comparator **234**b, and the comparator **234**b determines whether the second power supply voltage VCC**2** and the power terminal voltage V.sub.T are similar to each other. Then, the enable signal SW**2**_Enable of the second switch is input to the NAND gate **235**b.
(72) The relationship between the second control voltage V.sub.C2 and the second power supply voltage V.sub.C2 may satisfy Equation 2 below.
$V$.sub.C2=$VCC$2+Δ$V$.sub.2 (Equation 2)
(73) As shown in Equation 2, the second control voltage V.sub.C2 may be set to a higher voltage than the second power supply voltage VCC**2** by a second excessive voltage ΔV.sub.2. Since the second control voltage V.sub.C2 is a voltage used to switch the second switch SW**2**, the second switch SW**2** can be sufficiently turned on. For example, when the second switch SW**2** is implemented as an N-type transistor, a voltage of the second switching driving signal V.sub.SW2 must be set higher than the second power supply voltage VCC**2** so that the second switch SW**2** can sufficiently perform a turn-on operation. Accordingly, the second control voltage generator **230**b generates a second control voltage V.sub.C2 that is higher than the second power supply voltage

VCC**2** and outputs it to the switch controller **220**.
(74) FIG. **8** shows an internal configuration of the switch circuit **210** and an internal configuration of the switch controller **220** according to the example.
(75) As shown in FIG. **8**, the switch controller **220** may include a logic circuit **221** and a buffer circuit **222**.
(76) The logic circuit **221** receives external bit signals bit**1** and bit**2**, and generates and outputs logic signals V.sub.LOG1 and V.sub.LOG2 in response to the bit signals bit**1** and bit**2**. The first bit signal bit**1** and the first logic signal V.sub.LOG1 are used to control the first switch SW**1**, and the second bit signal bit**2** and the second logic signal V.sub.LOG2 are used to control the second switch SW**2**.
(77) FIG. **9** shows the logic circuit **221** according to the example.
(78) As shown in FIG. **9**, the logic circuit **221** may include a first NAND gate **910**, a second NAND gate **920**, and a third NAND gate **930**.
(79) The first NAND gate **910** receives the first bit signal bit**1** and the second bit signal bit**2**. The second NAND gate **920** receives the first bit signal bit**1** and the output of the first NAND gate **910**, and outputs the first logic signal V.sub.LOG1. Then, the third NAND gate **930** receives the second bit signal bit**2** and the output of the first NAND gate **910**, and outputs the second logic signal V.sub.LOG2.
(80) FIG. **10** shows an input-output logic table of the logic circuit **221** of FIG. **9**.
(81) As shown in FIG. **10**, the logic circuit **221** may generate and output four states in response to two bit signals. When the first logic signal V.sub.LOG1 is 1, it corresponds to a high level, and in this case, the first switch SW**1** may be turned off. In addition, when the first logic signal V.sub.LOG1 is 0, it corresponds to a low level, in this case, the first switch SW**1** may be turned on. Accordingly, the first and second switches SW**1** and SW**2** are turned off at the high level of the first and second logic signals V.sub.LOG1 and V.sub.LOG2, and the first and second switches SW**1** and SW**2** are turned on at the low level of the first and second logic signals V.sub.LOG1 and V.sub.LOG2. That is, since the first logic signal V.sub.LOG1 in bit signal **01** is at a low level, the first switch SW**1** is turned on. In addition, in bit signal **10**, since the second logic signal V.sub.LOG2 is at a low level, the second switch SW**2** is turned on. In the remaining cases, since the first and second logic signals V.sub.LOG1 and V.sub.LOG2 are at a high level, both the first and second switches SW**1** and SW**2** are turned off.
(82) The buffer circuit **222** may receive the first and second logic signals V.sub.LOG1 and V.sub.LOG2 from the logic circuit **221**, and may generate and output the switching driving signals V.sub.SW1 and V.sub.SW2. The buffer circuit **222** converts the first logic signal V.sub.LOG1 into the first switching driving signal V.sub.SW1 and converts the second logic signal V.sub.LOG2 into the second switching driving signal V.sub.SW2. Since the first logic signal V.sub.LOG1 and the second logic signal V.sub.LOG2 are logic signals, the current level is low. Accordingly, the buffer circuit **222** converts the first logic signal V.sub.LOG1 and the second logic signal V.sub.LOG2 into the first switching driving signal V.sub.SW1 and the second switching driving signal V.sub.SW2 having a high current level, respectively. The buffer circuit **222** may further include a level shifter circuit as well as a buffer to increase a voltage level as well as a current level. The buffer circuit **222** receives the first control voltage V.sub.C1 from the control voltage generator **230** and generates a first switching driving signal V.sub.SW1 having a high voltage level by using the first control voltage V.sub.C1. When the first switching driving signal V.sub.SW1 has a high voltage level, the first switching driving signal V.sub.SW1 may be the first control voltage V.sub.C1. The buffer circuit **222** receives the second control voltage V.sub.C2 from the control voltage generator **230** and generates a second switching driving signal V.sub.SW2 having a high voltage level by using the second control voltage V.sub.C2. When the second switching driving signal V.sub.SW2 has a high voltage level, the second switching driving signal V.sub.SW2 may be the second control voltage V.sub.C2. In addition, when the first and second driving signals V.sub.SW1 and V.sub.SW2

have a low voltage level, the first and second driving signals V.sub.SW1 and V.sub.SW2 may have a ground voltage of 0 V or a negative (−) voltage. When the first and second switches SW**1** and SW**2** are implemented as N-type transistors, the buffer circuit **222** is an ON driving signal, and the first and second switching driving signals V.sub.SW1 and V.sub.SW2 each having first and second control voltages V.sub.C1 and V.sub.C2, respectively, can be output. A method for the buffer circuit **222** to generate the first and second switching driving signals V.sub.SW1 and V.sub.SW2 using the first and second control voltages V.sub.C1 and V.sub.C2 may be a conventional method.
(83) As shown in FIG. **8**, in the switch circuit **210**, the first and second switches SW**1** and SW**2** may be respectively implemented as transistors **211** and **212**. As an example, the transistor **211** may be implemented as an N-type field effect transistor (FET).
(84) A first terminal (e.g., a drain terminal) of the transistor **211** is connected to the first power supply circuit **100***a* to receive the first power supply voltage VCC**1** as input (supply), and a second terminal (e.g., a source terminal) of the transistor **211** is connected to the power supply terminal T_VCC. A control terminal (e.g., a gate terminal) of the transistor **211** receives the first switching driving signal V.sub.SW1 from the buffer circuit **222**. The first terminal of the transistor **212** is connected to the second power supply circuit **100***b* to receive (supply) the second power supply voltage VCC**2**, and the second terminal of the transistor **212** is connected to the power supply terminal T_VCC. The control terminal of the transistor **212** receives the second switching driving signal V.sub.SW2 from the buffer circuit **222**.
(85) When the transistor **211** is an N-type transistor, the buffer circuit **220** may output a first control voltage V.sub.C1 as the first switching driving signal V.sub.SW1 for turn-on of the transistor **211**. In addition, when the transistor **212** is an N-type transistor, the buffer circuit **220** may output a second control voltage V.sub.C2 as the second switching driving signal V.sub.SW2 for turn-on of the transistor **212**. Through this, the N-type transistors **211** and **212** may be sufficiently turned on. When the control voltage of the N-type transistors **211** and **212** is lower than the voltage of the first terminal (e.g., the drain voltage) or the voltage of the second terminal (e.g., the source voltage), the N-type transistors **211** and **212** operated in a turn-off region such that ON resistance may be increased. To solve such a problem, the control voltage generator **230** generates first and second control voltages V.sub.C1 and V.sub.C2 that are higher than the first and second power supply voltages VCC**1** and VCC**2**, respectively. The buffer circuit **222** generates first and second control voltages V.sub.C1 and V.sub.C2 as control voltages of the N-type transistors **211** and **212**, respectively. That is, the first control voltage V.sub.C1 may be an ON control voltage of the N-type transistor **211**, and the second control voltage V.sub.C2 may be an ON control voltage of the N-type transistor **212**.
(86) While this disclosure includes specific examples, it will be apparent to one of ordinary skill in the art that various changes in form and details may be made in these examples without departing from the spirit and scope of the claims and their equivalents. The examples described herein are to be considered in a descriptive sense only, and not for purposes of limitation. Descriptions of features or aspects in each example are to be considered as being applicable to similar features or aspects in other examples. Suitable results may be achieved if the described techniques are performed to have a different order, and/or if components in a described system, architecture, device, or circuit are combined in a different manner, and/or replaced or supplemented by other components or their equivalents. Therefore, the scope of the disclosure is defined not by the detailed description, but by the claims and their equivalents, and all variations within the scope of the claims and their equivalents are to be construed as being included in the disclosure.

## Claims

1. A power supply switch circuit comprising: a first switch configured to switch supply of a first power supply voltage to a power supply terminal of a power amplifier; a control voltage generator

configured to generate a first control voltage through a charge pump operation, and to control the charge pump operation by comparing the first power supply voltage to a first voltage, which is a voltage of the power supply terminal; and a switch controller configured to generate a switching driving signal to control the first switch using the first control voltage.
2. The power supply switch circuit of claim 1, further comprising a second switch configured to switch supply of a second power supply voltage to the power supply terminal, wherein the control voltage generator is configured to generate a second control voltage through a second charge pump operation, and to control the second charge pump operation by comparing the second power supply voltage to the first voltage, and the switch controller is configured to generate a second switching driving signal to control the second switch using the second control voltage.
3. The power supply switch circuit of claim 1, wherein the control voltage generator comprises: a comparator configured to compare the first power supply voltage to the first voltage; an oscillator configured to generate a waveform corresponding to an enable signal of the first switch and an output of the comparator; and a charge pump configured to receive the first power supply voltage and a second voltage, and to generate the first control voltage by operating with the waveform.
4. The power supply switch circuit of claim 3, further comprising a counter configured to count a number of clocks of the waveform, and to output an enable signal to the comparator when the number of clocks satisfies a reference number, wherein the comparator is configured to determine whether a difference between the first power supply voltage and the first voltage is included within a set range when the enable signal of the counter is input.
5. The power supply switch circuit of claim 4, wherein the oscillator is configured to stop generating the waveform when the comparator determines that the difference between the first power supply voltage and the first voltage is included within the set range.
6. The power supply switch circuit of claim 4, wherein the control voltage generator further comprises a NAND gate configured to receive the enable signal of the first switch and an output of the comparator, and the oscillator is activated or deactivated corresponding to an output signal of the NAND gate.
7. The power supply switch circuit of claim 3, wherein the first control voltage corresponds to a sum of the first power supply voltage and the second voltage.
8. The power supply switch circuit of claim 1, wherein the switch controller comprises a buffer circuit configured to receive the first control voltage and to generate the switching driving signal having the first control voltage.
9. The power supply switch circuit of claim 1, wherein the first power supply voltage is fluctuated according to an envelope of a radio signal (RF) input to the power amplifier.
10. An operation method of a power supply switch circuit that supplies at least one of a first power supply voltage and a second power supply voltage to a power supply terminal of a power amplifier by switching the at least one voltage, the method comprising: generating a first waveform corresponding to an enable signal with respect to a first switch that switches the first power supply voltage; generating a first control voltage by performing a charge pump operation through the first waveform; comparing the first power supply voltage to a first voltage, which is a voltage of the power supply terminal; stopping the generating of the first waveform corresponding to a result of the comparing; and generating a first switching driving signal that controls the first switch using the first control voltage.
11. The operation method of claim 10, further comprising counting a number of clocks with respect to the first waveform, wherein the comparing comprises comparing the first power supply voltage to the first voltage when the number of clocks satisfies a reference number.
12. The operation method of claim 10, wherein the stopping comprises stopping generating the first waveform when a difference between the first power supply voltage and the first voltage is included within a set range.
13. The operation method of claim 10, further comprising: generating a second waveform

corresponding to an enable signal with respect to a second switch that switches the second power supply voltage; generating a second control voltage by performing a second charge pump operation through the second waveform; comparing the second power supply voltage to the first voltage; stopping generating the second waveform corresponding to a result of the comparing between the second power supply voltage and the first voltage; and generating a second switching driving signal that controls the second switch using the second control voltage.

14. The operation method of claim 13, wherein the first control voltage is higher than the first power supply voltage, and the second control voltage is higher than the second power supply voltage.

15. The operation method of claim 13, wherein the first power supply voltage and the second power supply voltage are fluctuated according to an envelope of a radio frequency (RF) signal input to the power amplifier.